Problem A. Status Code

Time Limit 2000 ms

Problem Statement

You are given an integer S between 100 and 999 (inclusive).

If S is between 200 and 299 (inclusive), print `Success`; otherwise, print `Failure`.

Constraints

- $100 \leq S \leq 999$
- S is an integer.

Input

The input is given from Standard Input in the following format:

```
S
```

Output

Print the answer.

Sample 1

Input	Output
200	Success

200 is between 200 and 299, so print `Success`.

Sample 2

Input	Output
401	Failure

401 is not between 200 and 299, so print `Failure`.

Sample 3

Input	Output
999	Failure

Problem B. ABC/ARC

Time Limit 2000 ms

Problem Statement

Smeke has decided to participate in AtCoder Beginner Contest (ABC) if his current rating is less than 1200, and participate in AtCoder Regular Contest (ARC) otherwise.

You are given Smeke's current rating, x. Print `ABC` if Smeke will participate in ABC, and print `ARC` otherwise.

Constraints

- $1 \leqq x \leqq 3{,}000$
- x is an integer.

Input

The input is given from Standard Input in the following format:

x

Output

Print the answer.

Sample 1

Input	Output
1000	ABC

Smeke's current rating is less than 1200, thus the output should be `ABC`.

Sample 2

Input	Output
2000	ARC

Smeke's current rating is not less than 1200, thus the output should be `ARC`.

Problem C. New Generation ABC

Time Limit 2000 ms

Problem Statement

This is the 214-th AtCoder Beginner Contest (ABC).

The ABCs so far have had the following number of problems.

- The 1-st through 125-th ABCs had 4 problems each.
- The 126-th through 211-th ABCs had 6 problems each.
- The 212-th through 214-th ABCs have 8 problems each.

Find the number of problems in the N-th ABC.

Constraints

- $1 \leq N \leq 214$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

N

Output

Print the answer.

Sample 1

Input	Output
214	8

Sample 2

Input	Output
1	4

Sample 3

Input	Output
126	6

Problem D. Wrong Answer

Time Limit 2000 ms

Problem Statement

You are given two integers A and B, each between 0 and 9, inclusive.

Print any integer between 0 and 9, inclusive, that is not equal to $A + B$.

Constraints

- $0 \leq A \leq 9$
- $0 \leq B \leq 9$
- $A + B \leq 9$
- A and B are integers.

Input

The input is given from Standard Input in the following format:

```
A B
```

Output

Print any integer between 0 and 9, inclusive, that is not equal to $A + B$.

Sample 1

Input	Output
2 5	2

When $A = 2, B = 5$, we have $A + B = 7$. Thus, printing any of $0, 1, 2, 3, 4, 5, 6, 8, 9$ is correct.

Sample 2

Input	Output
0 0	9

Sample 3

Input	Output
7 1	4

Problem E. Beginner

Time Limit 2000 ms

Problem Statement

Takahashi is a member of a programming competition site, *ButCoder*.

Each member of ButCoder is assigned two values: **Inner Rating** and **Displayed Rating**.

The Displayed Rating of a member is equal to their Inner Rating if the member has participated in 10 or more contests. Otherwise, the Displayed Rating will be their Inner Rating minus $100 \times (10 - K)$ when the member has participated in K contests.

Takahashi has participated in N contests, and his Displayed Rating is R. Find his Inner Rating.

Constraints

- All values in input are integers.
- $1 \leq N \leq 100$
- $0 \leq R \leq 4111$

Input

Input is given from Standard Input in the following format:

N R

Output

Print his Inner Rating.

Sample 1

Input	Output
2 2919	3719

Takahashi has participated in 2 contests, which is less than 10, so his Displayed Rating is his Inner Rating minus $100 \times (10 - 2) = 800$.

Thus, Takahashi's Inner Rating is $2919 + 800 = 3719$.

Sample 2

Input	Output
22 3051	3051

Problem F. New Scheme

Time Limit 2000 ms

Problem Statement

Given eight integers $S_1, S_2, \ldots,$ and S_8, print `Yes` if they satisfy all of the following three conditions, and `No` otherwise.

- The sequence $(S_1, S_2, \ldots, S_8)$ is monotonically non-decreasing. In other words, $S_1 \leq S_2 \leq \cdots \leq S_8$.
- $S_1, S_2, \ldots,$ and S_8 are all between 100 and 675, inclusive.
- $S_1, S_2, \ldots,$ and S_8 are all multiples of 25.

Constraints

- $0 \leq S_i \leq 1000$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

S_1 S_2 $\ldots$ S_8

Output

Print the answer.

Sample 1

Input	Output
125 175 250 300 400 525 600 650	Yes

They satisfy all of the three conditions.

Sample 2

Input	Output
100 250 300 400 325 575 625 675	No

They violate the first condition because $S_4 > S_5$.

Sample 3

Input	Output
0 23 24 145 301 413 631 632	No

They violate the second and third conditions.